

Haptic Force-Cepx: A Real-Time Simulation and Grip Feedback Device for Surgical Training

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Abstract—Current robot-assisted surgery platforms, such as the Da Vinci 5, lack grip feedback, increasing the risk of tissue slippage and damage. In this project, we present the Haptic Force-Cepx, a novel real-time simulation and grip feedback device for robot assisted surgery training. The system couples a low-impedance, direct-drive BLDC motor and magnetic encoder with a computer simulation of soft tissue dynamics. We present a two-loop control architecture to link the device to the simulation, with the outer simulation loop estimates tool–tissue interaction forces and fits a time-varying spring–damper model from recent frames, while the inner loop on the device renders the resulting forces to the user in real time. The simulation is able to render the visual environment and haptic parameters at 100-200 Hz, while the spring-damper model renders haptic feedback at over 1000 Hz, reducing instabilities in the system and providing a smooth user experience.

Video Link: [Click Here](#)

I. INTRODUCTION

Nowadays, Robot-assisted surgery (RAS) is increasingly being introduced into hospitals for its greater precision and stability. However, since many of these systems are teleoperated, the surgeon loses the inherent haptic feedback compared with holding the tools in their hands. For example, the current da Vinci 5 Surgical System does not have any force feedback when the user squeezes the forceps. This lack of force feedback can cause difficulties applying the correct grip force, leading to either slipping of tissue due to too little grip force applied, or the damage of delicate tissue due to too much grip force applied.

Prior work also shows that force feedback on gripper can measurably improve both surgical safety and training, which motivates our project. In [1] the authors develop a force-feedback technology for robotic-assisted surgery and show that surgeons apply significantly lower tissue forces when feedback is present, reducing trauma and improving patient outcomes. In [2] the authors build a 4-DOF telemanipulator with both translational and grip force feedback and find that only the grip force feedback helps users maintain forces near the no-slip threshold, highlighting the specific importance of grip-force cues. In [3] the authors create a force-detecting neurosurgical gripper with strain gauges and force reflection, enabling surgeons to feel small gripping and tensile forces when removing deep-seated tumors. Finally, in [4] the authors review haptic simulators and conclude that combining haptic

devices with physically based soft-tissue models substantially improves surgical training performance.

Therefore, we want to build a surgical training system to help surgeons become familiar with the mechanical properties of different human tissues during their use of the Da Vinci robot. In this paper, we present a motor-driven gripper connected to a simulation of the surgical environment where the system can simulate the interaction between gripper and tissue and compute the reaction force in real time. Then this force can be transmitted back to the gripper, enabling surgeons to perceive realistic grip forces.

II. METHODS

A. Device Description

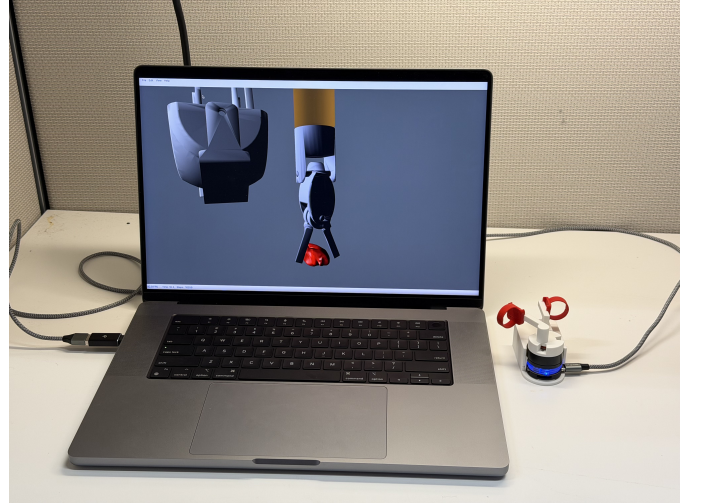


Fig. 1. Grip control device with simulation

The device is a kinesthetic haptic device designed to stimulate the muscle and joint receptors in the hand to create the feeling of gripping soft tissue. The completed prototype can be seen in Fig. 1. The following paragraphs detail its two core components: the physical grip controller and the computational simulation.

The physical device is built off of the LILYGO T-Knob (Shenzhen Xinyuan Electronic Technology Co., China), which contains an ESP32-C6-MINI-1U connected to a brushless direct current (BLDC) motor, a TMC6300 motor driver chip and an MT6701 magnetic encoder. The motor is used in a direct-drive configuration with a 3D-printed lever to create the haptic

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grip controller. Using a direct drive mechanism eliminates many of the mechanical losses that reduce backdriveability and increase the impedance of the device in free space.

The BLDC motor allows for much higher torque than a brushed motor of the same size, but it is more complex to control since torque is not controlled directly by sending a voltage to the motor. To solve this, functions included from LILYGO based off of the SimpleFOC library [5] are used to control the torque output of the motor.

B. Simulation Components

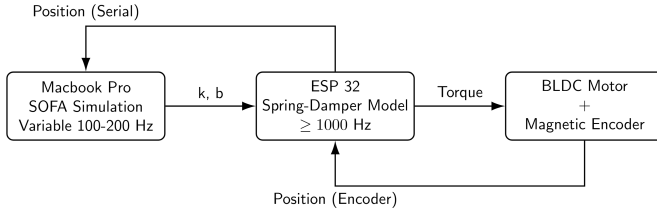


Fig. 2. Block diagram

1) **Object Interaction:** The key issue with real-time force feedback in simulation is the discrepancy between the lower frequency of the physics simulation (100-200Hz) due to hardware limit and the high frequency required for quality haptic feedback (≥ 1000 Hz). To remedy this, an inner haptic loop and outer simulation loop architecture is used as shown in Fig. 2. The inner haptic loop uses an ESP32 to render a virtual linear spring-damper model

$$F = -k(x - x_0) - b\dot{x} \quad (1)$$

updating the force of the motor at higher frequency. The outer simulation loop then sends updated values of the parameters, k , b , and x_0 to the ESP32 at a lower frequency to capture nonlinear tissue behaviors. Although this slightly decreases the accuracy of the force felt by the user, it greatly improves stability and user experience.

2) **Stiffness Estimation:** The local stiffness of the virtual tissue is calculated from the differential change in contact force F_k and device position $x_{device,k}$ at step k respectively. The raw stiffness estimate $K_{raw,k}$ is computed using a finite difference approach:

$$K_{raw,k} = \left| \frac{\Delta F}{\Delta x_{device}} \right| = \left| \frac{F_k - F_{k-1}}{x_{device,k} - x_{device,k-1}} \right| \quad (2)$$

To prevent numerical instability or division by zero, Eq. (2) is evaluated only when the contact force exceeds a threshold $F_k > \epsilon_F$ and the displacement $|\Delta x| > \epsilon_x$.

The direct use of K_{raw} can lead to strong vibrations due to the variable discrete time step. We apply an auto-regressive filter on stiffness $\hat{K}_k$:

$$\hat{K}_k = \alpha K_{raw} + (1 - \alpha) \hat{K}_{k-1} \quad (3)$$

where $\alpha \in [0, 1]$ is the smoothing factor. The smoothed $\hat{K}_k$ is a weighted average of past frames.

3) **Damping Estimation:** The damping coefficient $\hat{B}_k$ is adapted to stiffness and a user-defined virtual mass m_v to release excess energy in the system:

$$\hat{B}_k = 2\sqrt{\hat{K}_k \cdot m_v} \quad (4)$$

When contact is lost (i.e., $F_k < \epsilon_F$), the stiffness is not immediately set to zero but is decayed by a factor λ_{decay} to ensure a smooth release of the gripper. User should feel nothing if leaving the spring range, and the stiffness parameter is not used.

$$\hat{K}_k = \lambda_{decay} \hat{K}_{k-1} \quad (5)$$

4) **Proxy Position Calculation:** The virtual proxy position $x_{proxy,k}$, which represents the equilibrium point of the local spring model at time k , is computed by inverse Hooke's equation:

$$x_{proxy,k} = x_{device,k} + \frac{F_k}{\hat{K}_k} \quad (6)$$

Finally, the parameter set $\{\hat{K}_k, \hat{B}_k, x_{proxy,k}\}$ is transmitted to the embedded controller to update the high-frequency impedance loop. Detailed logic is at Alg. 1

5) **Force Simulation:** The contact force F_k become the only value need from simulator. Multi-object force simulation in 3-D space involves complex non-linear boundary analysis. To achieve real-time performance for the interaction between the PSM and the liver. We utilize proxy spheres as shape representation of the surface geometry during deformation, as shown in Fig 3, which is more real-time computable compared to performing collision detection on the all volumetric meshes. The soft tissue behavior is modeled using Finite Element Method (FEM) [4], and deformation is modeled using linear elasticity [6].

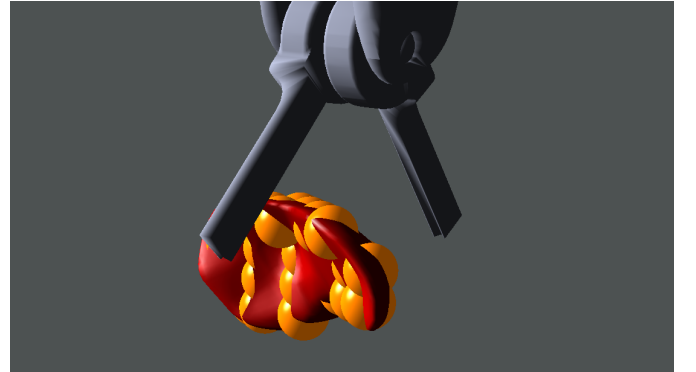


Fig. 3. Tissue deformation proxy spheres

The contact response is handled by a constraint solver where non-penetration and Coulomb friction are formulated as a Linear Complementarity Problem (LCP) [7]. The solver provided by SOFA computes Lagrange multipliers (λ), by an energy-based method, to satisfy these dynamical constraints. The total

magnitude of forces is the approximate force on gripper, and is then processed to render haptic feedback. Detailed logic is in Alg. 2

6) **Motor Control:** Render the spring-damper model, Alg. 3

By our observation, when an object is squeezed, the surface may vibrate due to sudden collision and get out of touch for a few frames, because of simulator's discrete timestep-based nature incapable to calculate the continuous real world. The unstable vibration can be scaled up and greatly influence user experience. As such, we filter out the "bouncing" if it lasts shorter than a time threshold ϵ_t .

To convert the calculated contact force F into the motor's torque control signal, we employ a non-linear mapping strategy. The mapping function, denoted as $M(x)$, acts as a dynamic gain coefficient that modulates the output torque based on the normalized force magnitude, where f is simulated contact force and f_{max} is a user-defined constant. The final motor torque is computed as:

$$\tau = \text{sgn}(\vec{F}) \cdot \tau_{max} \cdot M\left(\frac{|\vec{F}|}{F_{max}}\right) \quad (7)$$

where τ_{max} represents the peak torque capability of the motor. The objective of $M(x)$ is to amplify sensitivity for weak interactions (providing a high gradient near zero) while preventing motor saturation.

We investigated two strategies for the coefficient function $M(x)$:

$$M_{log}(x) = \frac{\ln(1+ax)}{\ln(1+a)}, \quad M_{tanh}(x) = \tanh(ax) \quad (8)$$

where a is a tunable scale strength parameter.

Both reinforce small force, which helps surgeon perception at the moment on contact, while having slight difference on the large force. Specifically, the logarithmic mapping M_{log} scales following a convex curve, providing high resolution for small forces. However, when $|\vec{F}| > F_{max}$, the force exceed normalization range, requiring an explicit hard clamp. In contrast, the hyperbolic tangent mapping M_{tanh} introduces a natural "soft normalization" behavior. This ensures the output torque lies within the physical limit, offering a smoother haptic experience at all force.

7) **Other Optimization:** All components in the control system are compiled C++, including multi-core memory-level operation. The serial communication is customize bit-wise protocol. Signal processing and communication is decoupled, allowing each to run on its full speed without blocking.

C. Preliminary Results

The goal of the mechanical design was to be able to render appropriate forces when the user is in contact with the tissue, while also having very low impedance when moving in free space. The motor used is able to generate 0.031 N-m of torque. The direct drive transmission keeps friction effects negligible, so at an approximate arm length of 39 mm, a maximum force

of 0.80 N can be delivered to the user. This is plenty for rendering the haptic sensations of soft tissue, though in future work it may need to be improved upon if the simulation needs to render harder objects (bones, ligaments, etc.). Because the transmission is direct-drive, and the motor has low cogging and friction effects, the free space has very low impedance. Overall the mechanical design meets the necessary goals, with some room for improvement on the force output.

The motor torque was mapped from simulated force with easily perceptible resolution on small force at the moment on and off contact, as shown in Fig 4. The instability was filtered. The system produces zero torque when gripper is static. Fig. 5 in the appendix details the internal computation.

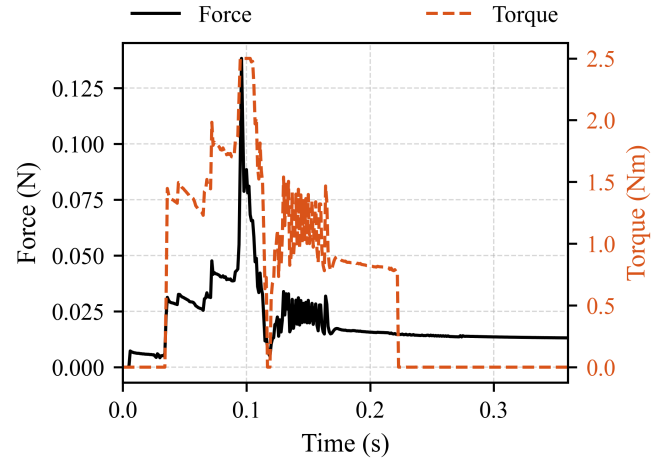


Fig. 4. Raw contact force from simulator and processed torque towards motor

III. FUTURE WORK

A. Future Development

A key future improvement is independent actuation of both gripper arms. On the current Da Vinci Surgical Console, each arm is controlled separately, enabling asymmetric grasps, rolling/scissoring motions, and finer control of contact forces than single-arm control allows. Our simulation already computes per-arm force feedback; the current prototype renders only their average due to its single-motor design, so the limitation is mechanical. Achieving true independence will require a second motor and encoder, a redesigned gripper mechanism, and a controller capable of running two motor control loops with dual encoder inputs.

Another future work would be connecting the grip device to additional degrees of freedom in order to implement force feedback in other directions. We would likely implement this by attaching the device to either the Da Vinci 5 console (which includes 3-DOF translational force feedback) or to another multi-DOF kinesthetic device such as the Sensable Phantom Premium. A similar algorithm using the same inner haptic

loop and outer simulation loop architecture would be used to implement further degrees of freedom.

B. User Study Protocol

A system performance study will be conducted to evaluate the accuracy and control efficacy of the Haptic force-cepx during remote simulated tissue manipulation tasks compared to traditional grippers without feedback. Around 20-30 participants will be recruited, including novice users, medical students, surgical trainees, and surgeons. During the experiment, all participants will operate the physical haptic force-cepx to complete a task required the user to grab a piece of tissue, and pull it to a target. Throughout the task, the user will be instructed to minimize the grip force exerted on the tissue. Participants will perform each task under different conditions—visual feedback only and visual feedback with haptic force feedback—following a randomized order to minimize learning bias. Each trial will be repeated 2-3 times for each task. All interaction data will be recorded for subsequent analysis of the haptic force-cepx performance evaluation.

C. Evaluation Metrics

A combination of quantitative metrics and qualitative metrics will be used to evaluate the task performance of the Haptic Force-Cepx. Task performance will be quantitatively evaluated using metrics such as distance to the target point, task completion time, applied force, the number of errors, and the area of tissue affected, which together reflect precision, effectiveness, and stability during remote simulated tissue manipulation. A post-experiment qualitative questionnaire will be administered to assess the subjective experience of the users. Qualitative feedback primarily evaluates the following aspects: the realism of tissue interactions, the intuitiveness of force feedback, the ability to discern soft-tissue boundaries, and overall satisfaction with the stability and responsiveness of force feedback.

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IV. APPENDIX

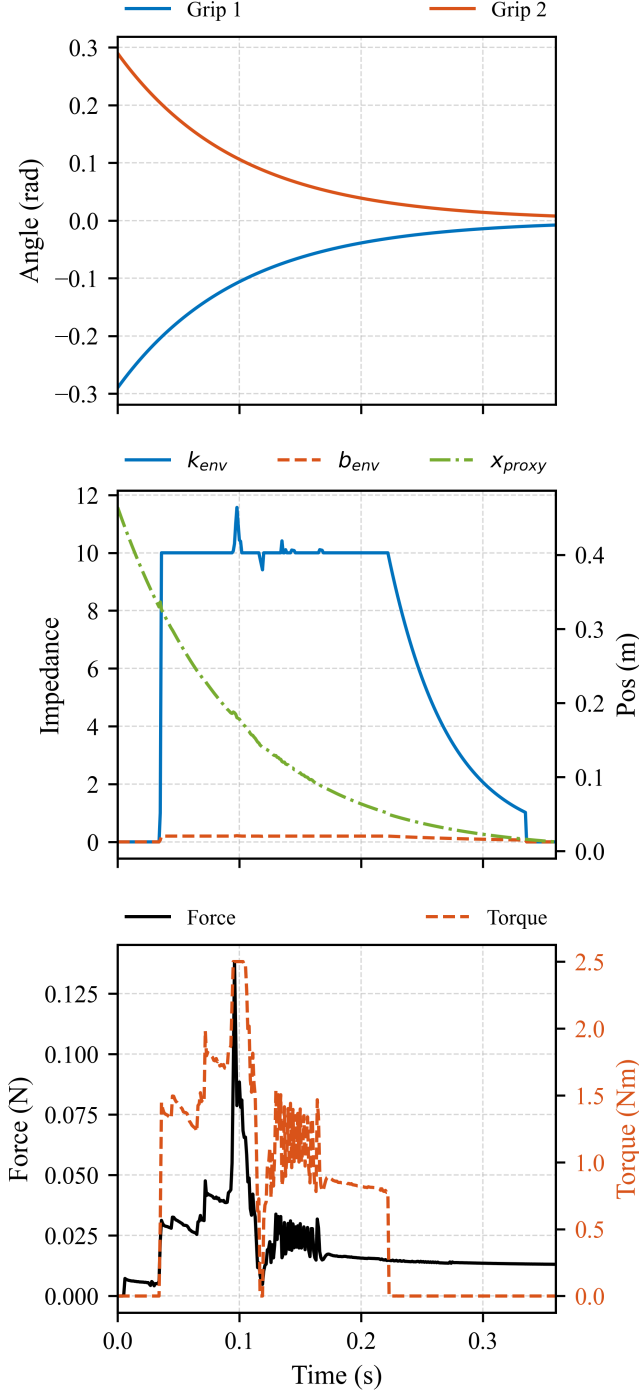


Fig. 5. Internal computation for motion that grippers squeeze a soft liver. Upper: gripper's angles during a pinch motion. Middle: estimated spring-damper parameters K, B, x_{proxy} . Bottom: raw contact force from simulator and processed torque towards motor

Algorithm 1 Simulation Loop on Computer

```

1: Initialize  $F_{prev} \leftarrow 0, x_{prev} \leftarrow 0, K \leftarrow 0$ , initial state  $S$ 
2: while Simulating do
3:    $x_{device} \leftarrow \text{READANGLE}$ 
4:    $S \leftarrow \text{STEP\_SIMULATION}(S, x_{device}, \Delta t)$ 
5:    $F \leftarrow \text{GETCONTACTFORCE}(S)$ 
6:    $\Delta F \leftarrow F - F_{prev}$ 
7:    $\Delta x \leftarrow x_{device} - x_{prev}$ 
8:   if  $F > \epsilon_{force}$  and  $|\Delta x| > \epsilon_{disp}$  then
9:      $K_{raw} \leftarrow |\Delta F / \Delta x|$ 
10:     $K \leftarrow \text{SMOOTHFILTER}(K_{raw})$ 
11:     $B \leftarrow 2\sqrt{K} \cdot m_{virtual}$ 
12:     $x_{proxy} \leftarrow x_{device} + (F/K)$ 
13:   else
14:      $K \leftarrow K \cdot \lambda_{decay}$ 
15:      $B \leftarrow 0$ 
16:   end if
17:    $\text{WRITEPARAM}(K, B, x_{proxy})$ 
18:    $F_{prev} \leftarrow F$ 
19:    $x_{prev} \leftarrow x_{device}$ 
20: end while

```

Algorithm 2 Contact Force

```

1: function  $\text{GETCONTACTFORCE}(\text{State } S)$ 
2:    $\vec{\lambda} \leftarrow \text{LCPSOLVER}(S)$ 
3:    $\vec{F}_{total} \leftarrow \vec{0}$ 
4:   for each constraint  $c$  do
5:      $\lambda_c \leftarrow \vec{\lambda}[c]$ 
6:      $\vec{n}_c \leftarrow \text{GETCONSTRAINTNORMALVEC}(c)$ 
7:     if  $\lambda_c > \epsilon$  then
8:        $\vec{F}_{total} \leftarrow \vec{F}_{total} + (\lambda_c \cdot \vec{n}_c)$ 
9:     end if
10:  end for
11:   $F \leftarrow \|\vec{F}_{total}\|$ 
12:  return  $F$ 
13: end function

```

Algorithm 3 Haptic Rendering Loop on ESP32

```

1: Initialize  $K \leftarrow 0, B \leftarrow 0, x_{proxy} \leftarrow 0$ 
2:  $t_{last} \leftarrow \text{GETTIME}$ 
3: while Running do
4:    $x_{device} \leftarrow \text{READENCODER}$ 
5:    $\dot{x}_{device} \leftarrow \text{COMPUTEVELOCITY}(x_{device})$ 
6:   if  $\text{NEWPARAMAVAILABLE}$  then
7:      $K, B, x_{proxy} \leftarrow \text{READPARAM}$ 
8:      $t_{last} \leftarrow \text{GETTIME}$ 
9:   end if
10:  if  $(\text{GETTIME} - t_{last}) > \epsilon_t$  then
11:     $K \leftarrow 0, B \leftarrow 0$ 
12:  end if
13:   $F \leftarrow K \cdot (x_{proxy} - x_{device}) - B \cdot \dot{x}_{device}$ 
14:   $\tau \leftarrow \text{MAPFORCETOTORQUE}(F)$ 
15:   $\text{WRITEMOTOR}(\tau)$ 
16:   $\text{SENDANGLE}(x_{device})$ 
17: end while

```
